



Databases and its Applications

Advanced NoSQL and Emerging Database trends

Pooja T S

Computer Applications

Databases and its Applications

Advanced NoSQL and Emerging Database trends

Types of NoSQL: Key-Value, Columnar, Graph, Document

Pooja T S

Computer Applications



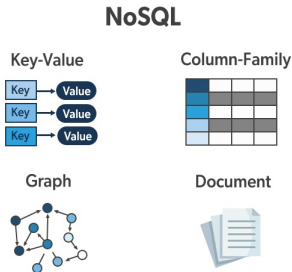
► Why classify NoSQL databases?

NoSQL databases are designed for different data models

Each type solves a specific class of problems

Choosing the right type improves performance and scalability

► Main NoSQL categories





Databases and its Applications

Key-Value Databases

► Core concept

Data stored as key-value pairs

Key is unique, value is opaque

Database does not understand the value structure

► Characteristics

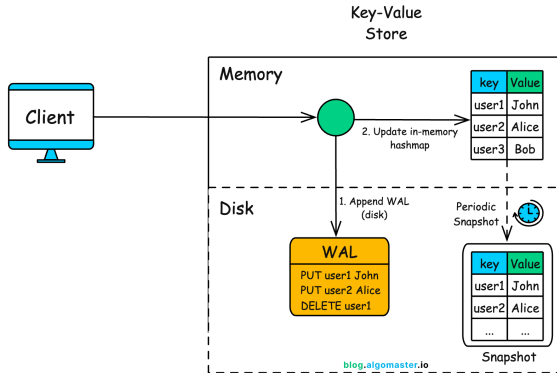
Extremely fast read and write operations

Simple data access model

Limited querying capabilities



Key-Value Databases





Databases and its Applications

Key-Value Databases – Use Cases

► Typical use cases

Caching (session data, user preferences)

Real-time recommendations

Shopping carts

► Popular examples

Redis

Amazon DynamoDB

Riak



Databases and its Applications

Columnar (Wide-Column) Databases

► Core concept

Data stored in columns instead of rows
Columns grouped into column families
Schema is flexible within families

► Key characteristics

High write throughput
Optimised for large-scale analytics
Excellent horizontal scalability



Databases and its Applications

Columnar Databases – Use Cases

► Typical use cases

- Time-series data
- Event logging
- Big data analytics

► Popular examples

- Apache Cassandra
- HBase
- Google Bigtable



Databases and its Applications

Graph Databases

► Core concept

Data represented as nodes and relationships

Relationships are first-class citizens

Optimised for traversals

► Key characteristics

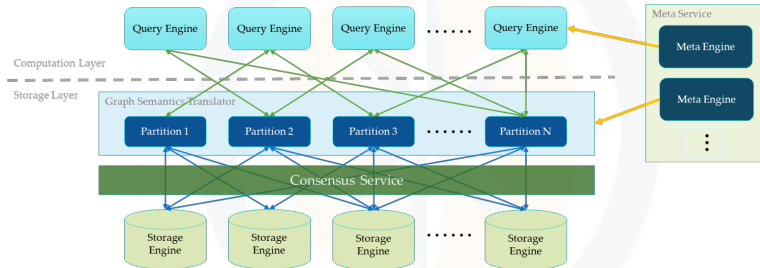
Fast relationship queries

Complex connections handled efficiently

Schema can evolve easily



Graph Databases





Databases and its Applications

Graph Databases – Use Cases

► Typical use cases

Social networks
Recommendation engines
Fraud detection

► Popular examples

Neo4j
Amazon Neptune
ArangoDB



Document Databases

► Core concept

Data stored as documents (JSON/BSON)

Each document is self-describing

Schema-less but structured

► Key characteristics

Rich querying support

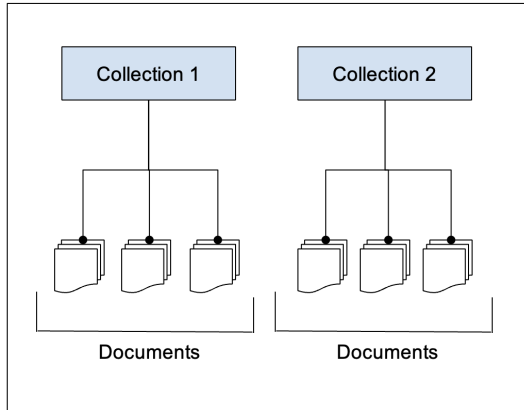
Nested and hierarchical data support

Easy mapping to application objects



Document Databases

Container (MongoDB Database)





Document Databases – Use Cases

► Typical use cases

Content management systems

E-commerce applications

Real-time analytics dashboards

► Popular examples

MongoDB

CouchDB

Firestore



Comparison and Summary of NoSQL Types

► **Comparison overview**

Key-Value: Speed and simplicity

Columnar: High-volume analytics

Graph: Relationship-intensive data

Document: Flexible and rich queries

► **Exam-focused keywords**

Key-value, column family, node-edge

BSON, document model

Horizontal scalability



PES
UNIVERSITY

CELEBRATING 50 YEARS

Pooja T S
Assistant Professor
Department of Computer Applications
poojats@pes.edu
080-26721983 Extn: 391